

EXPERIMENT 30: PROLOG PROGRAM FOR FORWARD CHAINING

Aim

To write and execute a Prolog program to demonstrate forward chaining.

Objective

- To understand forward chaining.
- To derive new information from existing facts.
- To apply rules repeatedly until a conclusion is obtained.
- To understand data-driven reasoning.

Algorithm

1. Define the initial facts.
2. Define rules for deriving new facts.
3. Start with the available facts.
4. Check each rule against the known facts.
5. If a rule's conditions are satisfied, derive its conclusion.
6. Add the new conclusion to the knowledge base.
7. Continue until no new facts can be derived.
8. Check the required query.

Flowchart

START

|

v

Enter Initial Facts

|

v

Check Rules

|

v

Are Conditions Satisfied?

/\

Yes No

| |

v v

Derive Check

New Fact Next Rule

|

v

Add Fact

|

v

More Rules?

/\

Yes No

| |

v v

Repeat Check Goal

|

v

STOP

Code

% Forward Chaining

fact(bird). fact(has_feathers).

rule(can_fly) :-

fact(bird),

```
fact(has_feathers).
```

```
rule(can_move) :-
```

```
    fact(bird).
```

```
query(can_fly) :-
```

```
    rule(can_fly).
```

Queries

```
?- fact(bird).
```

Output

```
true.
```

Query:

```
?- rule(can_fly).
```

Output:

```
true.
```

Query:

```
?- query(can_fly).
```

Output:

```
true.
```

Result

The program successfully demonstrates forward chaining by deriving conclusions from known facts.

Conclusion

Thus, forward chaining was successfully demonstrated using Prolog facts and rules.

```
users / varshini m / OneDrive / Desktop / Here!!! / A  
% Forward Chaining  
  
fact(bird).  
fact(has_feathers).  
  
rule(can_fly) :-  
    fact(bird),  
    fact(has_feathers).  
  
rule(can_move) :-  
    fact(bird).  
  
query(can_fly) :-  
    rule(can_fly).  
|
```

```
ERROR: Unknown procedure: fact/1 (SWI-Prolog)  
12 ?- make.  
% c:/users/varshini m/onedrive/desktop  
  exps/programs26 compiled 0.00 sec, -7  
true.  
  
13 ?- fact(bird).  
true.  
  
14 ?- query(can_fly).  
true.  
  
15 ?- rule(can_move).  
true.  
  
16 ?- |
```